Mahdi Kallel

PhD candidate in reinforcement learning

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Profile

- Graduating from the University of Würzburg in April 2027, advised by Carlo D'Eramo. I started out designing algorithms for exploration and policy optimization, and now study the learning dynamics of neural networks, in particular memory and plasticity in continual learning, and how reinforcement learning changes them compared with supervised learning.

Publications

The Spectral Anatomy of Plasticity Loss 2026
Uses random matrix theory to explain loss of plasticity, and shows that protecting the directions that hold old knowledge keeps networks trainable.
Mahdi Kallel, Nour Jamoussi, Johannes Tölle, Marios Kountouris, Carlo D'Eramo
openreview.net/forum?id=Kgz0zlJVS6 (AXIOM Workshop at NeurIPS 2026)

Do Not Imitate, Reinforce: Iterative Classification via Belief Refinement 2026
Trains classifiers by reinforcement learning, rewarding every step that improves the prediction. Achieves a better balance of accuracy and calibration than cross-entropy and stays robust to label noise.
Mahdi Kallel, Johannes Tölle, Ahmed Hendawy, Carlo D'Eramo
arxiv.org/abs/2604.22110 (Transactions on Machine Learning Research (TMLR), 2026)

PPO+: Enhancing Proximal Policy Optimization 2026
Shows that PPO is held back by its critic, and fixes it with an off-policy critic.
Mahdi Kallel, Jose-Luis Holgado-Alvarez, Samuele Tosatto, Carlo D'Eramo
rlj.cs.umass.edu/2026/papers/Paper8.html (Reinforcement Learning Journal (RLC 2026))

Use the Online Network If You Can: Towards Fast and Stable Reinforcement Learning 2026
Ahmed Hendawy, Henrik Metternich, Théo Vincent, **Mahdi Kallel**, Jan Peters, Carlo D'Eramo
arxiv.org/abs/2510.02590 (ICLR 2026)

Augmented Bayesian Policy Search 2024
Turns a learned critic into the prior of Bayesian policy search, outperforming earlier Bayesian optimization methods in high dimensions.
Mahdi Kallel, Debabrota Basu, Riad Akrour, Carlo D'Eramo
proceedings.iclr.cc/paper_files/paper/2024/hash/7eca17ef54789b0663cab421f2e9dbf5-Abstract-Conference.html (ICLR 2024)

Cyclicity-Regularized Coordination Graphs 2024
Oliver Järnefelt, **Mahdi Kallel**, Carlo D'Eramo
rlj.cs.umass.edu/2024/papers/RLJ_RLC_2024_40.pdf (RLC 2024)

Research Experience

INRIA Lille, SCOOL Research Group, Research Intern France
- Started the work on Bayesian policy search with deterministic policies that became Augmented Bayesian Policy Search (ICLR 2024). 2022

EURECOM, Research Intern — Biot, France, 2020
- Compared centralized and decentralized approaches to mobile traffic prediction and load balancing.

Industry Experience

Mcovision (independent), Machine Learning Engineer — Paris, France, 2021
- Built a classifier of echocardiography videos for cardiac amyloidosis and other rare heart diseases with Hôpital Cochin, from data cleaning to training.

SustainEcho, Research Intern — Paris, France, 2021
- Explored few-shot and hierarchical learning methods for classification with little data.

Education

PhD **University of Würzburg**, Reinforcement Learning — Germany, 2023 – 2027
- Expected April 2027. Supervisor: Prof. Carlo D'Eramo.
- Teaching: Introduction to Machine Learning; Introduction to Reinforcement Learning.

MSc **Université Paris-Saclay**, Optimization — France, 2021 – 2022
- Coursework: Convex analysis, dynamic programming, stochastic optimization and approximation.

MSc **ENS Paris-Saclay**, Mathematics, Vision, Learning (MVA) — France, 2020 – 2021
- Coursework: Reinforcement learning, convex optimization, kernel methods, advanced deep learning.

BSc **Télécom Paris / EURECOM**, Data Science — France, 2018 – 2020

Supervision, Talks and Service

Supervision: Johannes Tölle, MSc thesis, University of Würzburg (2026). The thesis received the VDE Bayern Award 2026, and Johannes co-authored two of the papers above.

Talks: On Better Priors for Policy Search. Invited talk and lab visit, Improbable AI Lab (Pulkit Agrawal), MIT, August 2024.

Reviewer: NeurIPS (top reviewer, 2026), ICLR, ICML, RLC

Technical Skills

Programming: Python, JAX, PyTorch, NumPy, Julia, C

Languages: Arabic (native), English (fluent), French (fluent), German (basic)